Pregnancy Risk Models - V2

Synthetic Training + Maternal Real-World Validation

Project Scope

Compare 4 models
Risk labels: low risk, mid risk, high risk
BodyTemp excluded in testing

Real Test Dataset

Rows used: 1014
Source: Maternal_Health_Risk_Data_Filled.csv
Features aligned to V2 model schema

Compared Models

- Random Forest
- Logistic Regression
- SVM
- XGBoost

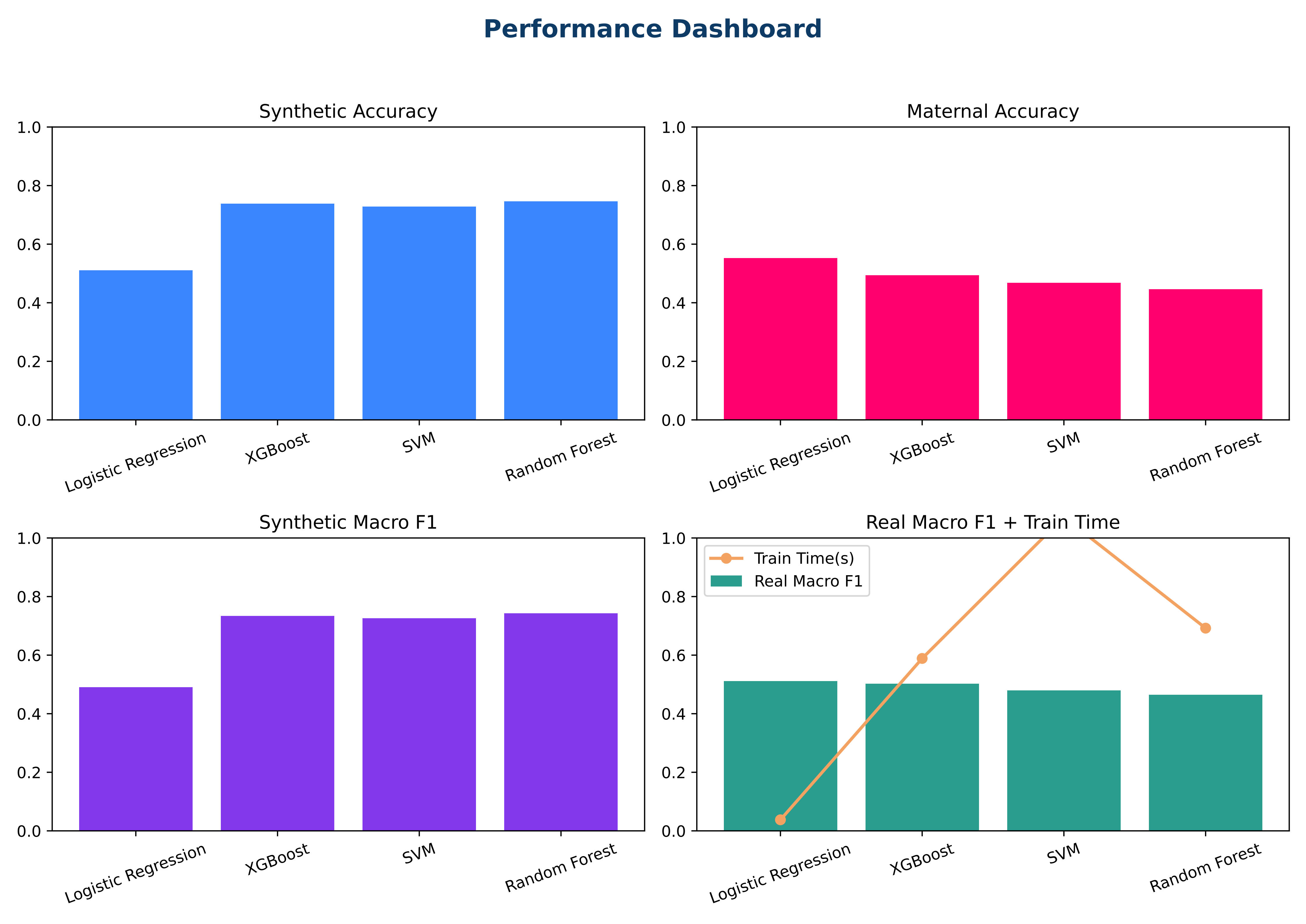
Metrics

- Accuracy and Macro F1
- F1 per class
- Training time
- Model strengths/weak points

Model Comparison Table

Model	Train(s)	Syn Acc	Real Acc	Real Macro F1	F1 HIGH	F1 LOW	F1 MEDIUM
ogistic Regressi	0.038	0.510	0.552	0.511	0.625	0.629	0.279
XGBoost	0.589	0.738	0.494	0.502	0.699	0.514	0.294
SVM	1.072	0.728	0.468	0.480	0.679	0.396	0.364
Random Forest	0.692	0.746	0.446	0.464	0.687	0.425	0.280

Best model on maternal testing: Logistic Regression


Performance Dashboard
Synthetic Accuracy
Maternal Accuracy
Synthetic Macro F1
Real Macro F1 + Train Time
Train Time(s)
Real Macro F1
Logistic Regression
XGBoost
SVM
Random Forest
0.0
0.2
0.4
0.6
0.8
1.0

Model Strengths and Weaknesses

Logistic Regression
Power points: Best F1 on LOW: 0.629 | Best recall on HIGH: 0.798 | Best precision on MEDIUM: 0.604
Weak points: Weakest F1 on MEDIUM: 0.279 | Weakest recall on MEDIUM: 0.181 | Weakest precision on HIGH: 0.514

XGBoost
Power points: Best F1 on HIGH: 0.699 | Best recall on HIGH: 0.710 | Best precision on HIGH: 0.689
Weak points: Weakest F1 on MEDIUM: 0.294 | Weakest recall on MEDIUM: 0.283 | Weakest precision on MEDIUM: 0.305

SVM
Power points: Best F1 on HIGH: 0.679 | Best recall on HIGH: 0.743 | Best precision on HIGH: 0.625
Weak points: Weakest F1 on MEDIUM: 0.364 | Weakest recall on LOW: 0.345 | Weakest precision on MEDIUM: 0.339

Random Forest
Power points: Best F1 on HIGH: 0.687 | Best recall on HIGH: 0.691 | Best precision on HIGH: 0.684
Weak points: Weakest F1 on MEDIUM: 0.280 | Weakest recall on MEDIUM: 0.292 | Weakest precision on MEDIUM: 0.269

Where Models Perform Well / Poorly

Logistic Regression
Good points: idx 127 true=HIGH pred=HIGH conf=1.00 ; idx 137 true=HIGH pred=HIGH conf=1.00
Not good points: idx 444 true=MEDIUM pred=HIGH conf=1.00 ; idx 797 true=MEDIUM pred=HIGH conf=1.00

XGBoost
Good points: idx 123 true=HIGH pred=HIGH conf=1.00 ; idx 130 true=HIGH pred=HIGH conf=1.00
Not good points: idx 505 true=LOW pred=HIGH conf=1.00 ; idx 910 true=LOW pred=HIGH conf=1.00

SVM
Good points: idx 436 true=HIGH pred=HIGH conf=1.00 ; idx 425 true=HIGH pred=HIGH conf=1.00
Not good points: idx 74 true=MEDIUM pred=HIGH conf=0.99 ; idx 331 true=MEDIUM pred=HIGH conf=0.99

Random Forest
Good points: idx 114 true=HIGH pred=HIGH conf=0.98 ; idx 502 true=HIGH pred=HIGH conf=0.98
Not good points: idx 413 true=MEDIUM pred=HIGH conf=0.95 ; idx 787 true=MEDIUM pred=HIGH conf=0.95

Final Recommendation

Conclusion

Best model on maternal testing: Logistic Regression.

This report uses maternal clinical rows with textual labels (low risk, mid risk, high risk), ignores BodyTemp during testing, and evaluates robustness across four models.

Recommendation:
- Keep the best model as deployment candidate
- Track weaker classes using class-level F1 shown above
- Use good/not-good test-point examples to guide future dataset enrichment